



Data Flow Diagrams

Blood Inventory Management System

Comprehensive DFD Analysis - Level 0 to Level 3



External Entity - Users/Systems outside the system



Process - System operations/transformations



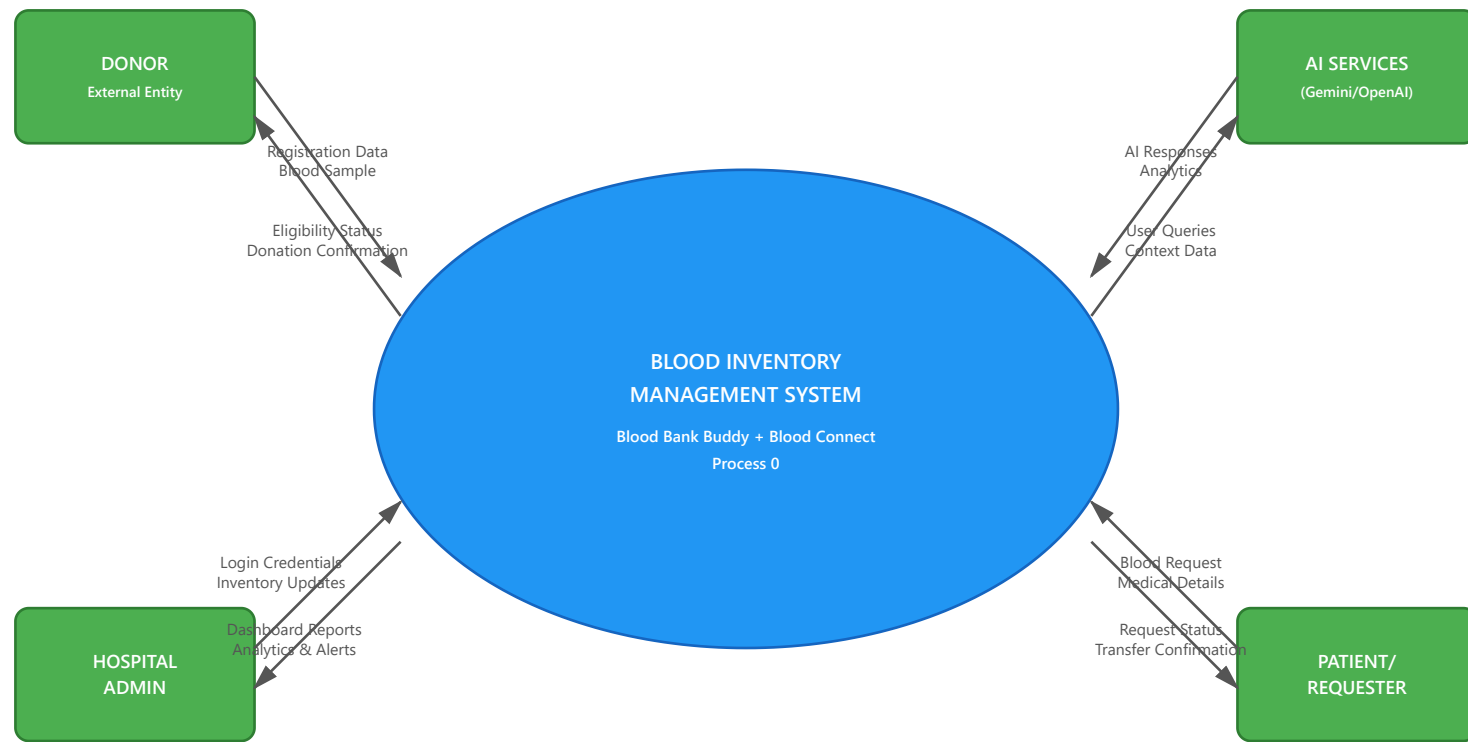
Data Store - Databases/Files



Sub-Process - Detailed process breakdown

Level 0 - Context Diagram

The highest level view showing the entire Blood Inventory Management System as a single process with external entities.



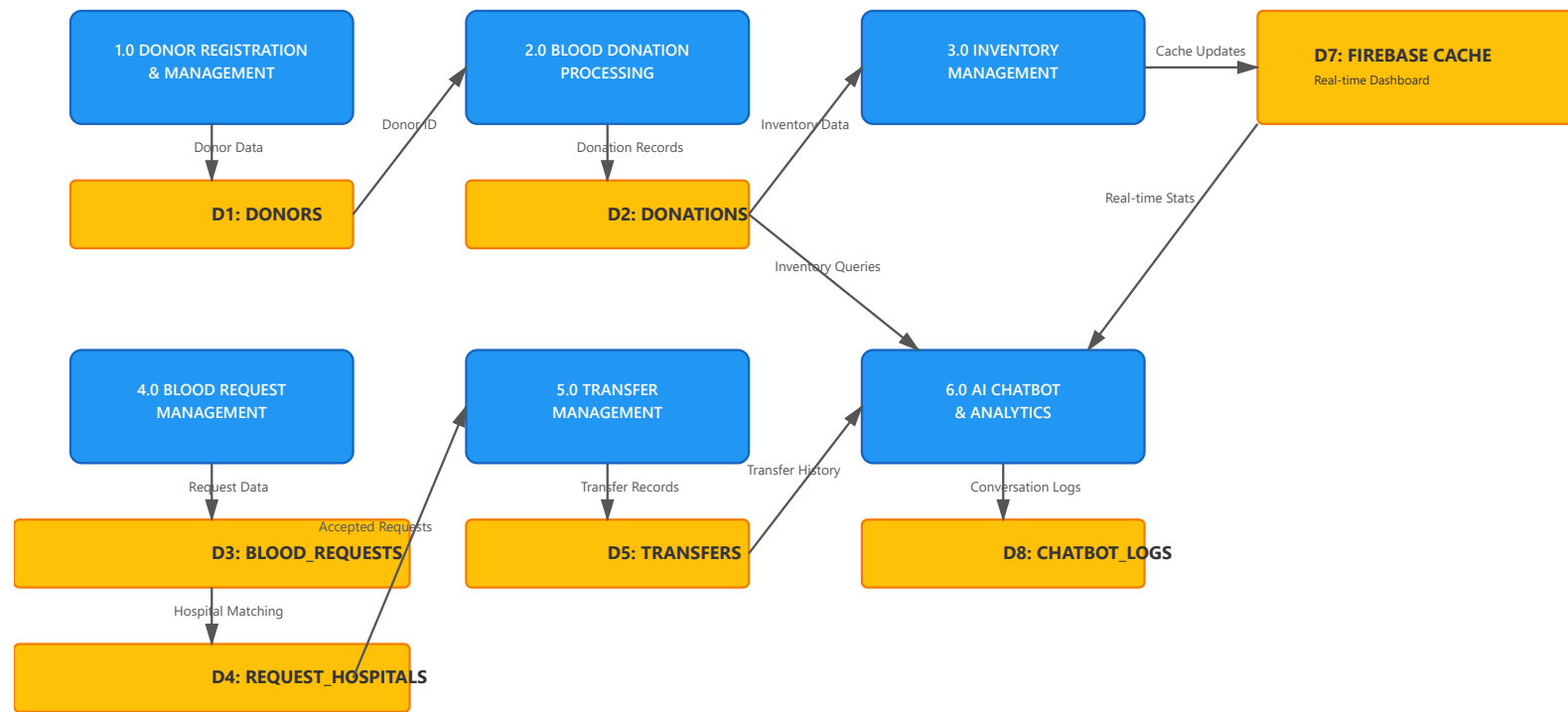
External Entities Overview

- **Donor:** Individuals donating blood, providing personal and medical information
- **Hospital Admin:** Hospital staff managing blood inventory and operations
- **Patient/Requester:** Individuals requesting blood for medical treatment

- **AI Services:** External AI APIs (Google Gemini, Azure OpenAI) for intelligent assistance

Level 1 - System Overview

Breaks down the main system into 6 Major Processes with 8 Data Stores



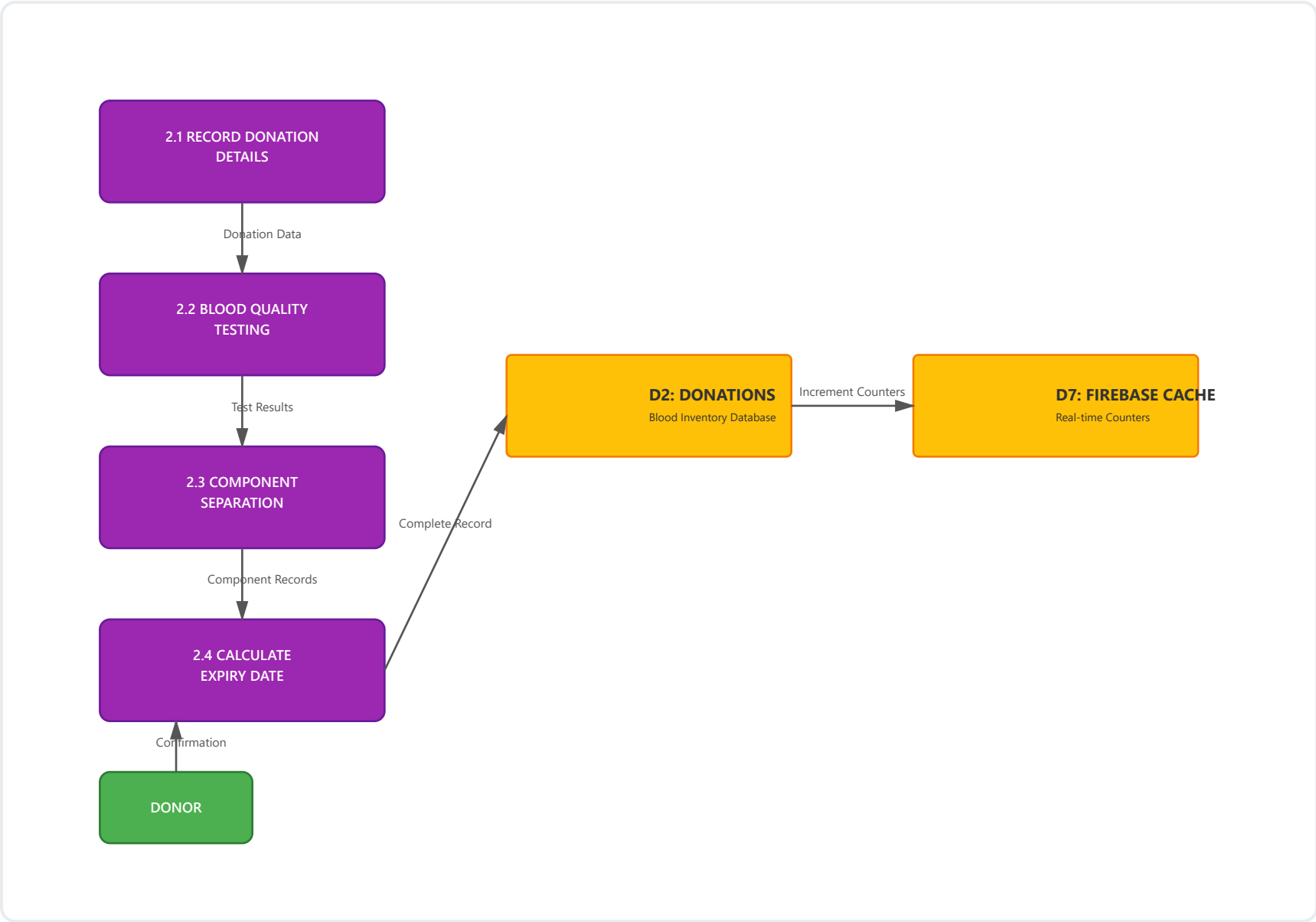
Major Processes Overview

Process ID	Process Name	Description	Key Functions
1.0	Donor Registration & Management	Capture and validate donor information	Registration, Eligibility Check, Profile Updates
2.0	Blood Donation Processing	Record donations and manage blood quality	Donation Recording, Quality Testing, Component Separation
3.0	Inventory Management	Track and monitor blood inventory	Stock Tracking, Expiry Monitoring, Alert Generation
4.0	Blood Request Management	Process blood requests from patients	Request Reception, Blood Matching, Hospital Notification
5.0	Transfer Management	Handle blood transfers to patients	Transfer Creation, Status Tracking, Delivery Confirmation
6.0	AI Chatbot & Analytics	Provide intelligent assistance and insights	Query Processing, Data Retrieval, Response Generation

Level 2 - Detailed Process Breakdown

Each major process is decomposed into sub-processes showing internal data flows.

Process 2.0: Blood Donation Processing (Detailed)



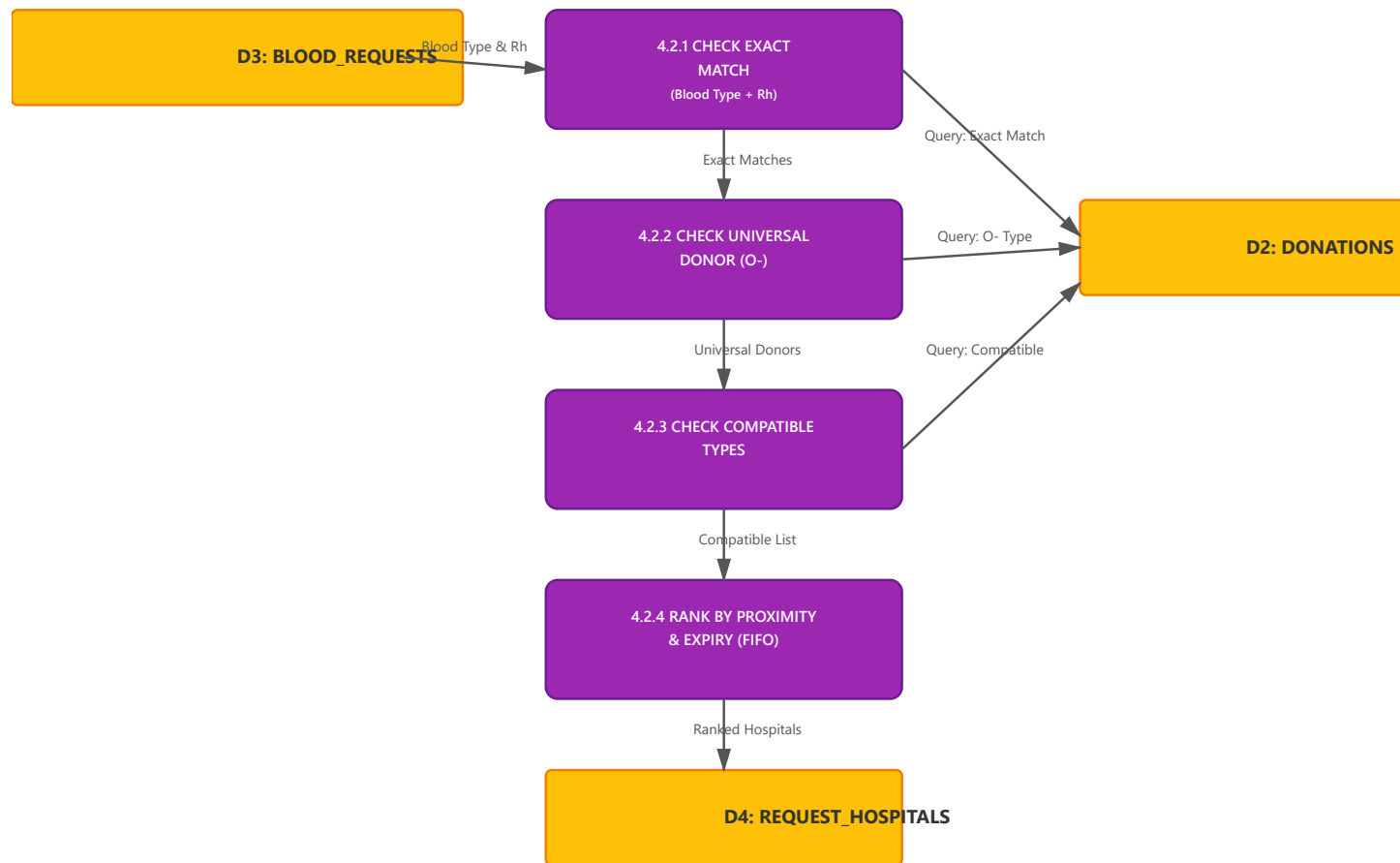
Process 4.0: Blood Request Management (Detailed)



Level 3 - Granular Sub-Process Details

Deep dive into complex operations showing implementation-level data flows.

Process 4.2: Match Blood Types (Detailed Implementation)



Blood Compatibility Rules

- **A+** can receive: A+, A-, O+, O-
- **A-** can receive: A-, O-

- **B+** can receive: B+, B-, O+, O-
- **B-** can receive: B-, O-
- **AB+** can receive: All blood types (Universal Recipient)
- **AB-** can receive: AB-, A-, B-, O-
- **O+** can receive: O+, O-
- **O-** can receive: O- only (Universal Donor)

Data Dictionary

Data Stores

ID	Name	Type	Contents	Primary Key
D1	DONORS	Azure SQL	Donor profiles and medical history	donor_id
D2	DONATIONS	Azure SQL	Blood inventory records	donation_id
D3	BLOOD_REQUESTS	Azure SQL	Patient blood requirements	request_id
D4	REQUEST_HOSPITALS	Azure SQL	Hospital notifications (Junction table)	request_id + hospital_id

ID	Name	Type	Contents	Primary Key
D5	TRANSFERS	Azure SQL	Blood transfer transactions	transfer_id
D6	HOSPITALS	Azure Cosmos DB	Hospital registry and authentication	hospital_id
D7	FIREBASE_CACHE	Firebase Realtime DB	Dashboard counters and real-time inventory	hospital_id
D8	CHATBOT_LOGS	Firebase Realtime DB	AI conversation history	session_id

Key Data Elements

Element Name	Data Type	Length	Description	Example
donor_id	INT	-	Unique donor identifier	1001
hospital_id	VARCHAR	10	Hospital unique code	"H001"
donation_id	INT	-	Blood unit identifier	5001
blood_type	VARCHAR	5	ABO blood group	"A+", "O-"
rh_factor	VARCHAR	1	Rhesus factor	"+", "-"

Element Name	Data Type	Length	Description	Example
volume_ml	INT	-	Blood volume in milliliters	450
urgency_level	VARCHAR	20	Request priority	"Critical", "High", "Medium", "Low"
status	VARCHAR	20	Current status	"Available", "Used", "Expired"
component_type	VARCHAR	50	Blood component	"Whole Blood", "Plasma", "Platelets"
is_eligible	BIT	-	Donor eligibility flag	1 (true), 0 (false)

Data Flow Rules & Constraints

1. Donor Registration Flow Rules

- **Rule 1.1:** Donor must provide valid blood type (A, B, AB, O) and Rh factor (+, -)
- **Rule 1.2:** Donor age must be between 18-65 years
- **Rule 1.3:** Duplicate email addresses are not allowed
- **Rule 1.4:** Eligibility verification runs automatically after registration
- **Rule 1.5:** Last donation date is checked (must be >90 days for whole blood)

2. Donation Processing Flow Rules

- **Rule 2.1:** Donor must be marked as eligible before donation
- **Rule 2.2:** Volume must be between 350-500 ml for whole blood
- **Rule 2.3:** All quality tests (HIV, HBsAg, HCV, Syphilis) must pass
- **Rule 2.4:** Expiry date is automatically calculated:
 - Whole Blood: +35 days
 - Plasma: +365 days
 - Platelets: +5 days
 - RBC: +42 days
- **Rule 2.5:** Firebase cache is updated atomically with SQL insert

3. Blood Request Flow Rules

- **Rule 4.1:** Request must specify exact blood type and Rh factor
- **Rule 4.2:** Critical urgency requests notify all matching hospitals
- **Rule 4.3:** Hospitals can only accept requests if inventory available
- **Rule 4.4:** Request status changes to "Fulfilled" after transfer completion
- **Rule 4.5:** FIFO (First In, First Out) principle for blood distribution

4. Transfer Management Flow Rules

- **Rule 5.1:** Transfer can only be created for "Available" blood units
- **Rule 5.2:** Blood unit status changes to "Used" upon transfer creation
- **Rule 5.3:** Transfer cannot be deleted, only cancelled
- **Rule 5.4:** Firebase cache decrements inventory atomically with transfer
- **Rule 5.5:** Complete audit trail maintained for all transfers

Data Flow Timing

Process	Trigger	Frequency	Response Time
Donor Registration	User action	On-demand	< 2 seconds
Blood Donation Processing	User action	On-demand	< 3 seconds
Inventory Update	Donation created	Real-time	< 500 ms
Expiry Monitoring	Scheduled	Daily (midnight)	Batch process
Blood Request Notification	Request created	Real-time	< 1 second
Transfer Creation	User action	On-demand	< 2 seconds
Firebase Cache Sync	Data modification	Real-time	< 100 ms

Process	Trigger	Frequency	Response Time
Dashboard Refresh	Page load / Auto-refresh	Every 30 seconds	< 50 ms (cache)
AI Chatbot Query	User message	On-demand	1-3 seconds

Summary

DFD Documentation Overview

- **Level 0:** Context diagram showing 4 external entities and the system boundary
- **Level 1:** System decomposition into 6 major processes and 8 data stores
- **Level 2:** Detailed sub-process breakdown showing internal workflows
- **Level 3:** Granular implementation details for complex operations
- **Total Processes:** 25+ processes and sub-processes documented
- **Total Data Flows:** 50+ data flows mapped
- **Database Architecture:** Hybrid RDBMS (Azure SQL) + NoSQL (Cosmos DB, Firebase)

Key System Features

-  Real-time inventory management with Firebase caching

- ☒ AI-powered chatbot with dual-mode operation
- ☒ Blood type compatibility matching algorithm
- ☒ Inter-hospital blood transfer coordination
- ☒ Automated expiry monitoring and alerts
- ☒ Complete audit trail for all transactions
- ☒ FIFO (First In, First Out) inventory distribution
- ☒ Multi-component blood separation tracking

Blood Inventory Management System - Data Flow Diagrams

Document Version: 1.0 | Created: January 2, 2026

© 2026 Blood Inventory Management Team. All Rights Reserved.